

Streaming Active Learning with Deep Neural Networks

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Advanced Machine Learning



Artificial Intelligence
for the Sciences

Introduction: Why Active Learning?

The Setting

In many real-world problems, unlabeled data is abundant but obtaining labels is expensive.

The Gap

Prior deep active learning methods assume offline access to the entire dataset. But in practice, data often arrives as a stream and decisions must be made **on the fly**.

⇒ This paper proposes **VeSSAL**: a streaming, committal, hyperparameter-free active learning algorithm.

The Challenges of Streaming Active Learning

How can we efficiently identify samples from a data stream for neural networks while respecting a maximum query rate?

Desirable Properties

A good batch active learner must balance:

- **Uncertainty:** query points where the model is least confident.
- **Diversity:** avoid redundant queries, two similar uncertain points carry the same information.
- **Committal:** ask the user for a label when a sample is encountered.
- **Evenly distributed:** queries spread evenly across the data stream to match a maximum query rate q

Why Existing Methods Fail in Streaming

- Popular methods (BADGE, Coreset) require **simultaneous access** to all unlabeled data $\Rightarrow$ unusable for very large or fractured datasets.
- Streaming demands **committal decisions**: accept or reject each sample *the moment it arrives*, without revisiting.
- The total number of future samples is **unknown**.

VeSSAL = **V**olume **S**ampling for **S**teaming **A**ctive **L**earning

Key Design Choices

- 1 **How to represent** unlabeled samples? → *Gradient embeddings* (last-layer gradients)
- 2 **How to select** diverse samples? → *Approximate volume sampling* (determinantal criterion)

Properties of VeSSAL

- Sees each unlabeled point only once
- Commits to a labeling decision immediately upon encountering each sample.
- Seamlessly trades off uncertainty and diversity via a single probabilistic criterion.
- Matches a desired query rate q (rather than a fixed batch size).

Step 1: Gradient Embeddings (Representation)

For a neural network f with last-layer parameters θ_L and cross-entropy loss ℓ :

$$g(x_t) = \frac{\partial}{\partial \theta_L} \ell(f(x_t; \theta), \hat{y}_t), \quad \hat{y}_t = \arg \max f(x_t; \theta)$$

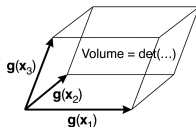
- **Uncertainty** $\leftrightarrow$ **gradient norm** $\|g(x)\|$: a confident prediction yields a small gradient; an uncertain prediction yields a large one.
- **Diversity** $\leftrightarrow$ **gradient direction**: different input regions induce gradients pointing in *different directions*.

Step 2: Volume Sampling (Selection)

Select batch B with probability $\propto$ the determinant:

$$p_B \propto \det \left(\sum_{x \in B} g(x) g(x)^\top \right)$$

Large when vectors are both **long** (uncertain) and **spread out** (diverse).



The Math Behind the Streaming Decisions

Problem: Volume sampling over a batch requires seeing all candidates. How to do it *one point at a time*?

From Batch to Streaming: Determinantal Contribution

When sample x_t arrives, accept it with probability proportional to its **marginal determinantal contribution** given items already selected:

$$p_t \propto \det\left(\hat{\Sigma}_t + g(x_t)g(x_t)^\top\right)$$

where $\hat{\Sigma}_t = \sum_{x \in B_t} g(x)g(x)^\top$ is the covariance of already-selected points in the batch.

Efficient Computation via the Matrix Determinant Lemma

$$p_t \propto \det(\hat{\Sigma}_t) \left(1 + g(x_t)^\top \hat{\Sigma}_t^{-1} g(x_t)\right) \text{ (Matrix Determinant Lemma)}$$

$$\propto \underbrace{g(x_t)^\top \hat{\Sigma}_t^{-1} g(x_t)}_{\text{Elliptical Potential}}$$

This quantity is large when $g(x_t)$ is *not well explained* by the directions already in the batch $\Rightarrow$ the new point adds genuine information.

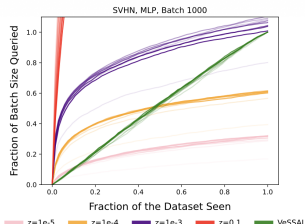
Autotuning the Query Rate

We need a scaling term z_t so that the expected query rate matches a target q :

$$\mathbb{E}_x[p_t] = \mathbb{E}_x\left[z_t \cdot g(x_t)^\top \hat{\Sigma}_t^{-1} g(x_t)\right] = q$$

Why a Fixed z_t Fails

The distribution of $g(x_t)^\top \hat{\Sigma}_t^{-1} g(x_t)$ changes every time a new point is added to $\hat{\Sigma}_t$. A fixed z_t leads to **drastic over- or under-sampling**.



Autotuning the scaling term

$$\begin{aligned}\mathbb{E}_x[p_t] &= z_t \cdot \text{tr}\left(\hat{\Sigma}_t^{-1} \mathbb{E}_x[g(x) g(x)^\top]\right) = q \\ \implies p_t &= q \cdot \frac{g(x_t)^\top \hat{\Sigma}_t^{-1} g(x_t)}{\text{tr}(\hat{\Sigma}_t^{-1} \frac{1}{t} \sum_{i=1}^t g(x_i) g(x_i)^\top)}\end{aligned}$$

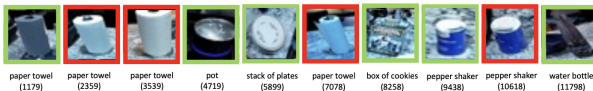
Now z_t is automatically determined $\Rightarrow$ hyperparameter-free!

$\hat{\Sigma}_t^{-1}$ updated efficiently via rank-1 Woodbury updates.

Experimental Results & Performance



(a) VeSSAL



(b) Uniform Rate Sampling

Key Findings

- **I.I.D. streams:** VeSSAL matches or nearly matches BADGE (best non-streaming method), despite the streaming constraint.
- **Non-I.I.D. streams** (adversarial sorting / natural drift): VeSSAL remains robust and still competitive with pool-based methods.
- **Diversity:** On CLOW, VeSSAL selects 10 objects from 9 different classes, while uniform sampling repeatedly picks the same class.
- **Compute:** VeSSAL's runtime stays nearly constant as batch size grows, while BADGE/Coreset scale superlinearly.

Summary

- VeSSAL is a streaming, committal, hyperparameter-free batch active learning algorithm for deep networks.
- Gradient embeddings jointly capture uncertainty (norm) and diversity (direction).
- Volume sampling with autotuned scaling distributes the labeling budget evenly across the stream.
- Competitive with state-of-the-art pool-based methods, with much lower computational cost.

Limitations

- Assumes the number of classes is known a priori, not realistic in open-world settings.
- The covariance estimate A_t can be biased under non-I.I.D. ordering (empirically not a major issue, but no theoretical guarantee).
- All queries are treated as equally costly, ignores variable annotation costs.
- Only evaluated on image classification; applicability to NLP or other modalities is unexplored.